7 Algorithm design and problem solving

1 Identify and describe **three** stages of the program development lifecycle.

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2 Abstraction, decomposition and structure diagrams are tools used during the program development lifecycle. Identify the stage of the program development lifecycle where they are used and describe what they are used for.

Abstraction

Stage: .....

Use: .....

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Decomposition

Stage: .....

Use: .....

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Structure diagram

Stage: .....

Use: .....

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3 a Identify **three** of the different items, other than software, that make up a computer system.

1 .....

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3 .....

b State **three** of the component parts of a computer system.

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4 Identify and describe the **three** methods you could use to design and construct a solution to a problem.

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5 Use the following list of words and phrases to complete the paragraph below.

Each word or phrase can be used once, more than once or not at all.

- breakdown
- bottom-up
- composition
- decomposition
- description
- diagrammatic
- flowchart
- formal
- hierarchical
- level
- program
- pseudocode
- top-down
- structure diagram
- system
- sub-systems

Structure diagrams are used to show the ..... design of a computer .....

in a ..... form.

A structure diagram shows the design of a computer program in a ..... way

with each ..... giving a more detailed ..... of the .....

into .....

6 a Draw and describe the use of **four** flowchart symbols.

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b Describe the purpose of flow lines in a flowchart.

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7 Describe the function of each of the following types of pseudocode statement and give an example in pseudocode of the use of each one.

Assignment:

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Conditional:

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Iterative:

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8 a There are **three** different types of loop structure in pseudocode.

For each one, describe the structure and write the pseudocode statements needed to output the names of 20 students that have been stored in the array `Student[0:19]`. Use the variable `Counter` for the iteration.

Loop 1: .....

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Loop 2: .....

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Loop 3: .....

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b i There are two different types of conditional statements used in pseudocode.

For each one, write the pseudocode statements needed to output a different welcome message for each name input from 'Alice', 'Abid', 'Dinesh', 'Daisy' or 'Zak' or a rejection message. Use the variable `Name` for the input.

Conditional Statements 1: .....

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Conditional Statements 2:

ii State, with reasons, which conditional statement is more appropriate to use in this case.

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c There are **two** different ways of writing a flowchart to output the names of twenty students that have been stored in the array `Student[0:19]`. The loop counter `Counter` is tested at the start of the loop (pre-condition) or at the end of the loop (post-condition).

i Draw flowcharts for the pre-condition loop and the post-condition loop.

Pre-condition loop:

Post-condition loop:

ii Identify which pseudocode loop statements matches each flowchart.

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d When you have learnt how to write programs that include loops, write three programs to test that your answers to part **a** work as intended.
Hint: for Python there will only be two programs.

Loop 1: .....

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Loop 2: .....

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Loop 3: .....

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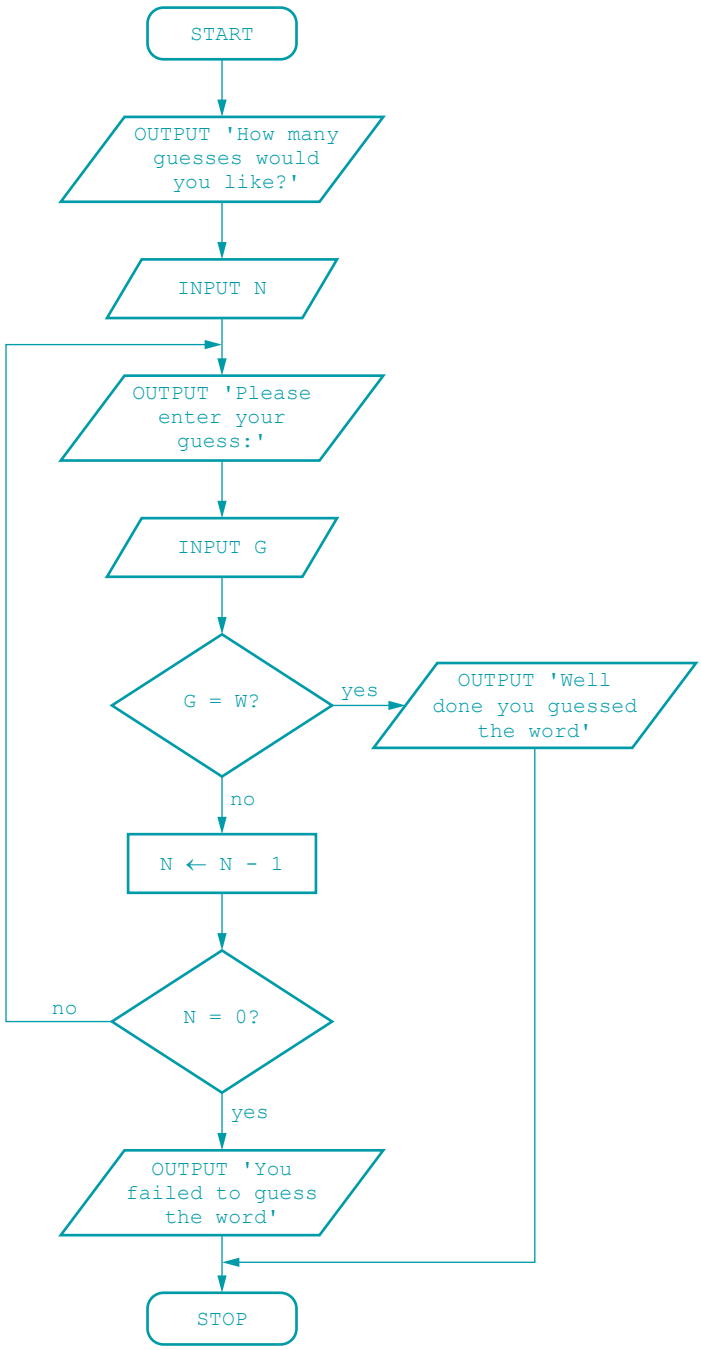
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9 a Write an algorithm in pseudocode to input ten numbers and output the total.

b Write an algorithm using a flowchart to input ten numbers and output the total.

10 Look at this algorithm, shown as a flowchart:



a Identify the processes in the algorithm.

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b Describe the purpose of the algorithm.

c Rewrite the algorithm in pseudocode.

11 Use the following list of words and phrases to complete the paragraph below.

Each word or phrase can be used once, more than once or not at all.

- algorithms
- average
- bubble
- counter
- counting
- linear
- list
- maximum
- mode
- minimum
- search
- sort
- testing
- totalling

Standard methods of solution are used in the design of ..... These include adding in a new value every time an action occurs, for example, awarding a mark to each student – this is called ..... When divided by the number of times this occurs, this gives the ..... Keeping a record of the number of times an action occurs is called ..... Finding the largest, ..., and smallest, ..., are also standard methods.

In order to look for an item in a list a ..... is used. The method you need to know for IGCSE Computer Science is to inspect each item in the ..... in turn to see if it is the one required. This is called a ....

To put a list in order a ..... is used. The method you need to know for IGCSE Computer Science is called a ....

12 a Explain the operation of a bubble sort.

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b Write a bubble sort algorithm in pseudocode to sort a list of ten names, in an array called `Names[]`, in ascending order.

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13 Validation and verification checks are used on values that are input into a computer program.

a i State when it is appropriate to use validation.

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ii State when it is appropriate to use verification.

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b i Give **two** examples of validation checks.

1 .....

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ii Give **two** examples of verification checks.

1 .....

2 .....

14 a Explain the purpose of test data.

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b i Boundary data is one type of test data. Identify and describe **three** other types of test data.

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ii Give the set of boundary data needed to check that a whole number is less than 100.

1 .....

2 .....

15 a Use the trace table and the test data 78, 34, 22, –4, 98, 16, 734, 88, 999 to record a dry run of this algorithm written in pseudocode.

```
W ← 0
X ← 0
Y ← 100
Z ← 0
REPEAT
     INPUT Mark
     IF Mark <> 999
       THEN
         REPEAT
              IF Mark < 0 OR Mark > 100
                THEN
                  INPUT Mark
              ENDIF
         UNTIL Mark >= 0 AND Mark <= 100
          IF Mark > X
           THEN
             X ← Mark
          ENDIF
          IF Mark < Y
           THEN
             Y ← Mark
         ENDIF
         Z ← Z + Mark
         W ← W + 1
     ENDIF
UNTIL Mark = 999
OUTPUT X, Y, Z
```

W	X	Y	Z	Mark	OUTPUT

b State the purpose of the algorithm.

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c Rewrite the algorithm as a flowchart.

d Use the guidance in Chapter 8 to rewrite this algorithm in pseudocode, ensuring that:
- there are meaningful prompts and outputs
- the algorithm could be read and understood by someone else.

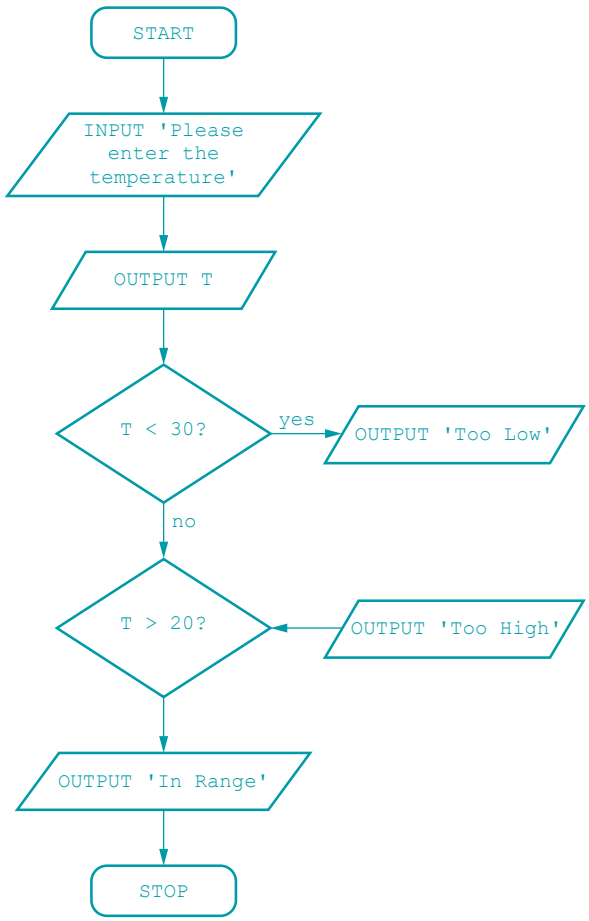
e Devise another set of test data that includes normal, erroneous and extreme data and complete a trace table to perform a dry run on your algorithm.

Test data: .....

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Trace table:

16 The following flowchart shows an algorithm that checks a temperature input is between 20 and 30 degrees. It contains several errors.



a Identify the errors in the algorithm and the flowchart construction.

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b Draw the corrected flowchart.